Apolipoprotein A-I_{Milano}

Correlation between High Density Lipoprotein Subclass Distribution and Triglyceridemia

Guido Franceschini, Laura Calabresi, Claudio Tosi, Cesare R. Sirtori, Claudia Fragiacomo, Giorgio Noseda, Elaine Gong, Patricia Blanche, and Alex V. Nichols

Carriers of the apolipoprotein A-I_{Milano} (apo A-I_M) variant represent a selected group of subjects showing low levels of high density lipoprotein (HDL), variable hypertriglyceridemia, and low prevalence of atherosclerotic vascular disease. The distribution of HDL subfractions and the correlation with abnormalities in triglyceride transport were determined in these subjects. Sera from 24 apo A-I_M carriers (A-I_M^+ and from age- and sex-matched normolipidemic controls (A-I_M^-) were analyzed by rate zonal ultracentrifugation. The A-I_M^+ subjects showed a marked decrease of HDL_3 mass with reduced flotation rates and major compositional alterations; the HDL_2 were nearly absent. The HDL subclasses from 10 A-I_M^+ subjects were resolved according to particle size by gradient gel electrophoresis (GGE). The HDL patterns detected in the carriers were unique in exhibiting a distinct peak in the $(HDL_{3b})_{GGE}$ interval, undetectable in the controls. Three patterns reflecting the relative contributions of smaller $(HDL_{3b})_{GGE}$ and larger $(HDL_{3a})_{GGE}$ particles could be distinguished in the carriers, and these were clearly related to different triglyceride and HDL cholesterol levels in plasma. These findings in a highly selected group of subjects with generally low HDL levels and quite variable triglyceridemia confirmed the existence of relationships between alterations in triglyceride transport and abnormalities in the HDL subclass distribution, possibly reflecting the variable atherosclerotic risk in hypertriglyceridemic subjects. (Arteriosclerosis 7:426–435, July/August 1987)

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A negative correlation between plasma high density lipoprotein (HDL) cholesterol level and triglyceridemia has been repeatedly demonstrated in epidemiological studies.[1] This correlation has been proposed as the basis for the association between hypertriglyceridemia and the risk of atherosclerotic vascular disease.[2] On the other hand, few data support a direct connection in selected groups of patients between an altered triglyceride (TG) transport and low HDL cholesterol (HDL-C) levels.[3]

Hypertriglyceridemia may frequently be present in disorders associated with low HDL levels,[4] such as Tangier and fish-eye diseases, as well as deficiencies of apo A-I–C-III and of lecithin-cholesterol acyltransferase (LCAT).[5] In some of these disorders, structurally and functionally abnormal HDL particles may account for the extremely variable prevalence of atherosclerotic disease in the affected subjects.[4]

A dramatic alteration in HDL apolipoprotein structure concomitant with hypertriglyceridemia has been detected in carriers of the apolipoprotein A-I_{Milano} (apo A-I_M) mutant, the first described molecular variant of human apolipoproteins.[6, 7] It is characterized by an Arg → Cys substitution in position 173 of the apo A-I sequence.[8] The presence of a cysteine residue in the variant apolipoprotein allows the formation of complexes with apo A-II, as well as of apo A-I_M, dimers.[7] In the originally identified carrier family, hypertriglyceridemia of variable degree was associated with a marked reduction of HDL-C with levels ranging between 7 and 18 mg/dl.[6] The HDL isolated from these carriers were TG-enriched and cholesteryl ester (CE)-depleted.[9] When pooled HDL from carriers were studied after cross-linking with dimethylsuberimidate,[10] a marked reduction of HDL_2 and a significant heterogeneity of HDL_3, characterized by the presence of three different subpopulations, could be detected. More recently, a large group of A-I_M^+ subjects have been uncovered in Limone sul Garda,[11] the village where the origin of the variant gene has been located. A significant elevation of mean plasma TG levels was

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observed in the A-$I_M$$^+$ group, compared to an age- and sex-matched group of controls. This was due in part to a significantly increased prevalence of hypertriglyceridemic subjects among the A-$I_M$$^+$. Both normolipidemic and hypertriglyceridemic A-$I_M$$^+$ subjects showed a significant TG enrichment in HDL, associated with a higher phospholipid (PL) and protein content, and with a marked reduction of CE.[12]

The A-$I_M$$^+$ individuals represent the most numerous known group of subjects carrying an apolipoprotein mutant, while showing a low prevalence of atherosclerotic vascular conditions.[11] In view of the still unclear relationship between HDL subfraction distribution, triglyceridemia, and atherosclerosis risk,[4] the determination of the HDL particle distribution in these subjects is of special interest. The HDL subfractions were resolved by two different techniques: according to lipoprotein flotation rates by rate zonal ultracentrifugation,[13] which separates subfractions by size, shape and density; and according to particle size, by nondenaturing polyacrylamide gradient gel electrophoresis (GGE).[14] It was thus possible to correlate the heterogeneity in the HDL pattern to abnormalities in plasma TG transport in the carriers of the mutant apolipoprotein.

Methods

Subjects

Analyses were carried out in 24 A-$I_M$$^+$ subjects (12 females and 12 males, mean age 38.5 ± 15.4 years) and 24 sex- and age-matched A-$I_M$$^-$ subjects from the described A-I_M kindred.[11] Eight A-$I_M$$^+$ subjects were hypertriglyceridemic (HTG), whereas the age- and sex-matched controls were all normolipidemic (NL).[12] Blood was collected after an overnight fast and serum was prepared by low speed centrifugation. EDTA (1 mM final concentration) and solid NaBr were immediately added to raise the nonprotein density to 1.40 g/ml; the sera were stored at 4° C.

Blood from ten A-$I_M$$^+$ subjects and from six A-$I_M$$^-$ subjects was also collected into EDTA (1 mg/ml) after an overnight fast; 1.4 mM 5,5′-dithiobis (2-nitrobenzoic acid) was added to plasma, which was immediately shipped on ice to Donner Laboratory, Berkeley, California. The procedures for HDL subfractionation were started within 48 hours after blood collection.

All subjects gave informed consent for the investigations.

Rate Zonal Ultracentrifugation

HDL subfractions were routinely isolated by a modification[15] of the rate-zonal density gradient ultracentrifugation procedure described by Groot et al.[16] After a 23-hour run at 40,000 rpm, the gradients were pumped through a Pharmacia (Uppsala, Sweden) UV monitor, and two fractions corresponding to HDL_3 and HDL_2 were collected with a Pharmacia FRAC-100 by pooling all material between the minimal optical density peaks recorded at 280 nm. The elution volume (Ve) of the HDL_3 and HDL_2 peaks and the total volumes of the two collected fractions were automatically monitored by the FRAC-100. The density profile along a "blank" gradient was measured at 15° C with an Abbe refractometer.

For apolipoprotein analyses, sera from three A-$I_M$$^+$ subjects were fractionated by the classical rate zonal ultracentrifugation (UCF) procedure in a Beckman Ti 14 rotor, as described by Patsch et al.[13] Fractions of 20 ml were collected, concentrated with an Amicon cell equipped with a PM 30 membrane (Amicon Corporation, Danvers, Massachusetts) and were analyzed for lipid and apolipoprotein composition after extensive dialysis against 0.15 M NaCl, 1 mM EDTA (pH 7.4).

Analytic Ultracentrifugation

Analytic ultracentrifugation and computer-derived Schlieren pattern analyses were carried out according to procedures described by Lindgren et al.[17]

Gradient Gel Electrophoresis

Ultracentrifugal d < 1.20 g/ml fractions were isolated from plasma and were analyzed by GGE, to determine particle size distribution of HDL. Nondenaturing electrophoresis using precast Pharmacia 4% to 30% slab gels (Pharmacia Fine Chemicals, Piscataway, New Jersey), was employed.[14] The reference proteins used to determine particle diameters were: thyroglobulin, apoferritin, lactate dehydrogenase, and bovine serum albumin. Densitometric scans of gradient gels were obtained with a Transidyne RFT densitometer (Transidyne General Corporation, Ann Arbor, Michigan).

Chemical Analyses

Total and free cholesterol (TC, FC), TG, and PL were determined by enzymatic procedures.[18, 19, 20] CE mass was calculated as (TC − FC) × 1.68.

The protein content of the HDL_2 and HDL_3 subfractions was evaluated by measuring the absorbances at 280 nm; the extinction coefficients determined in our laboratory for each subfraction were: 0.693 mg protein × A_{280} for HDL_2 and 0.782 mg protein × A_{280} for HDL_3. Protein in the Ti 14 fractions was determined by the method of Lowry et al.[21] Total lipoprotein mass was calculated as the sum of the individual components (FC + CE + PL + TG + P).

Isoelectric focusing (IEF) and SDS-electrophoresis in polyacrylamide gels were carried out on delipidated fractions as previously described.[9]

Results

Separation of HDL Subfractions by Rate Zonal Ultracentrifugation

Sera from all 48 selected subjects were analyzed by rate zonal ultracentrifugation in an SW 41 rotor. Each ultracentrifugal run processed sera from three A-$I_M$$^+$ subjects and from their respective age- and sex-matched A-$I_M$$^-$ controls. Two blank gradients were included every three runs to monitor the density profile of the gradient.

Ultracentrifugation of sera from the control subjects always resulted in a bimodal distribution of the 280 nm absorbance (Figure 1). Two fractions, designated as HDL_3

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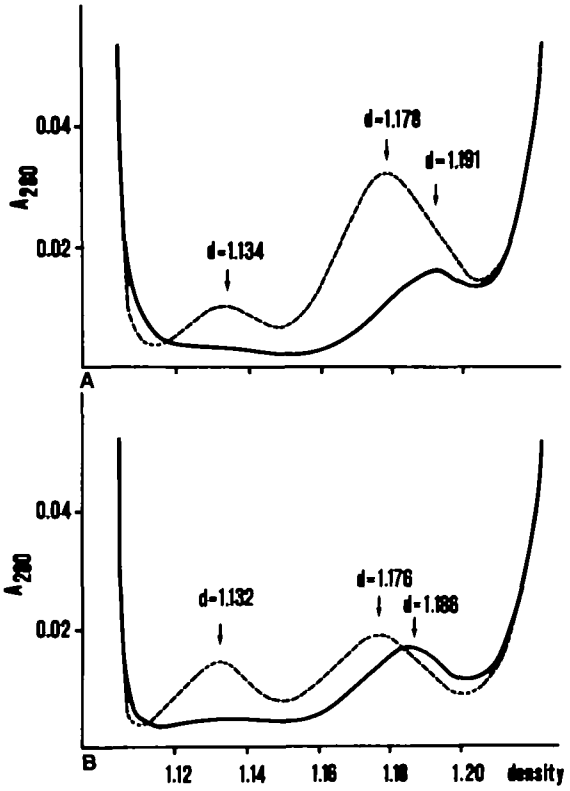


Figure 1. Rate zonal ultracentrifugation (SW41 rotor) profiles of HDL separated from two A-I_M^+ subjects (—) and from two A-I_M^- age- and sex-matched controls (--); **A.** Males. **B.** Females. Subjects can be identified from the reported kindred position.[11, 12] A-I_M^+: VI-171 (**A**), VII-201 (**B**); A-I_M^-: VI-129 (**A**), VII-138 (**B**).

and HDL_2, were collected; the Ve of the HDL_3 and HDL_2 peaks ranged between 3.60 and 4.53 ml (mean ± SD 4.03 ± 0.22 ml) and 6.55 to 7.48 ml (mean ± SD 7.05 ± 0.18), respectively.

When sera from A-I_M^+ subjects were analyzed, a different UV profile was generally observed (Figure 1). The HDL_3 eluted at a lower volume (range 2.73 to 4.03 ml) with a highly significant difference from A-I_M^- in the mean Ve (3.31 ± 0.36 ml; $p < 0.001$). Furthermore, a distinct HDL_2 peak was usually absent; in only seven A-I_M^+ subjects (four females and three males) was a rise of the UV absorbance beyond the HDL_3 tail detected. In these cases, the Ve for the apparent peak was in the normal range (6.98 ± 0.25 ml) for HDL_2. In the other 17 A-I_M^+ subjects, a fraction defined as "HDL_2" was collected between the end of the HDL_3 tail and the beginning of the low density lipoprotein (LDL) + very low density lipoprotein (VLDL) elution.

Slower floating HDL_3 particles could be detected in the 8 HTG A-I_M^+ subjects (Ve = 3.21 ± 0.34 ml), compared to NL A-I_M^+ (Ve = 3.42 ± 0.22 ml), whereas no differences were observed in HDL_2, a relatively distinct HDL_2 peak being detectable in three HTG and four NL A-I_M^+ subjects.

The A-I_M^+ subjects showed a marked reduction of both HDL_2 and HDL_3 when compared to A-I_M^- in terms of associated TC and total lipoprotein mass (Table 1). This was particularly true for the HDL_2 subfraction, the HDL_2/HDL_3 ratio being significantly lower in the A-I_M^+ subjects compared to the controls, both for TC (0.200 ± 0.069 vs. 0.435 ± 0.281; $p < 0.005$) and for lipoprotein mass (0.195 ± 0.064 vs. 0.337 ± 0.212; $p < 0.005$). The HTG A-I_M^+ subjects showed slightly lower TC levels compared to NL A-I_M^+ subjects, both in HDL_2 (3.0 ± 1.6 vs. 3.3 ± 1.9 mg/dl) and in HDL_3 (14.7 ± 6.1 vs. 16.7 ± 6.7 mg/dl), whereas the lipoprotein masses and ratios were identical in the two groups. In the controls, but not in the A-I_M^+ subjects, a significant difference could be detected in the HDL_2 levels between males and females (Table 1).

The composition of the HDL subfractions was markedly different in the A-I_M^+ subjects compared to A-I_M^- subjects; both HDL_2 and HDL_3 were TG- and protein-enriched but poor in CE (Table 2). In all the A-I_M^+ subjects, the composition of HDL_2 was very similar to that of HDL_3, thus confirming that in most of these subjects, an HDL_2 subfraction of clearly defined morphology and composition cannot be isolated.

To analyze the apolipoprotein composition of different lipoproteins in the HDL density range, sera from three NL A-I_M^+ subjects were subjected to UCF in a zonal Ti14 rotor. The material eluting in the HDL region was subdivided into 20 ml fractions and then analyzed for lipid and apolipoprotein composition (Figure 2). There were no significant variations in the lipid composition of particles eluting at different densities (data not shown).

The distribution of the three major A-I isoproteins along the elution profile of zonal UCF did not vary markedly

Table 1. Plasma Levels of Total Cholesterol and Mass of High Density Lipoprotein Subfractions

		HDL_2		HDL_3	
Subjects	No	TC	Mass	TC	Mass
Controls					
Total	24	15.8 ± 9.5	85.9 ± 46.7	38.1 ± 9.3	254.0 ± 58.3
Men	12	11.6 ± 5.3	60.6 ± 26.7	40.4 ± 12.4	274.2 ± 70.0
Women	12	20.0 ± 11.1*	110.1 ± 52.2†	35.8 ± 4.2	254.0 ± 46.9
A-I_M^+ subjects					
Total	24	3.2 ± 1.8‡	29.6 ± 10.4‡	16.0 ± 6.5‡	151.8 ± 57.8‡
Men	12	2.5 ± 1.5‡	25.6 ± 4.6‡	14.1 ± 6.0‡	140.4 ± 44.1‡
Women	12	3.9 ± 1.8‡	31.6 ± 10.9‡	17.8 ± 6.7‡	163.4 ± 62.3‡

All values represent the mean ± SD in mg/dl.
*$p < 0.05$ vs. men; †$p < 0.001$ vs. men; ‡$p < 0.001$ vs. controls.
TC = total cholesterol.

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Table 2. Percentage Lipid and Protein Composition of High Density Lipoprotein Subfractions

HDL	No	Free cholesterol	Cholesteryl ester	Phospholipid	Triglyceride	Protein
HDL_2						
A-I_M^-	24	3.4±1.2	25.2±2.8	26.4±3.9	3.5±1.1	41.5±4.7
A-I_M^+	7	1.9±1.3*	17.0±1.9†	27.2±3.5	5.9±1.5†	47.8±54.7*
A-I_M^+-NL	16	1.9±1.0	15.4±4.3	29.0±5.0	5.0±1.2	48.8±6.6
A-I_M^+-HTG	8	1.9±1.1	14.1±3.6	28.3±3.9	6.6±1.2‡	48.9±4.1
"HDL_2" A-I_M^+	17	1.9±0.9†	13.9±3.3†	30.0±4.6	5.5±1.5†	48.6±5.6†
HDL_3						
A-I_M^-	24	2.2±0.2	20.9±2.6	25.6±2.5	2.9±1.0	48.3±4.7
A-I_M^+	24	2.2±0.6	13.1±2.9†	26.6±2.4	5.0±1.4†	53.1±3.4†
A-I_M^+-NL	16	2.1±0.4	14.0±3.0	25.9±2.4	4.4±1.2	53.6±3.7
A-I_M^+-HTG	8	2.5±0.8	11.4±2.2‡	27.9±1.7‡	6.2±1.1§	52.0±2.4

All values represent the mean ± SD.
"HDL_2" identifies a fraction collected between the end of the HDL_3 tail and the beginning of the LDL + VLDL elution in the 17 A-I_M^+ subjects who did not show a distinct HDL_2 peak. HDL = high density lipoprotein.
*$p < 0.01$ vs. A-I_M^-; †$p < 0.001$ vs. A-I_M^-; ‡$p < 0.05$ vs. A-I_M^+-NL; §$p < 0.005$ vs. A-I_M^+-NL.

(Table 3). However, a slightly higher content of apo A-I-1, previously identified as a normal apo A-I in the A-I_M^+ subjects,[8] was noted in the ascending portion of the HDL_3 peak (fractions M through P), corresponding to HDL_3 in normals. A relative enrichment in apo A-I_M was observed both in fractions corresponding to HDL_2 in normals (J through K), as well as in lower flotation rate fractions where lipoproteins are not normally present (Q through U). The distribution of the different molecular forms of the apo A-I_M variant[7] along the HDL flotation profile was determined by SDS-polyacrylamide gel electrophoresis in nonreducing conditions. Both the monomeric apo A-I_M and the apo A-I_M–A-I_M and apo A-I_M–A-II dimers were present in all the zonal fractions, but their relative content varied widely. Two relatively homogeneous HDL populations were identified according to their content of these forms of apo A-I_M. Fractions F through P were enriched in the monomeric form (40% to 45% of total apolipoproteins), while the slower floating Q through T fractions contained a large amount (50%) of the apo A-I_M–A-II complex. The apo A-I_M–A-I_M dimer was present in variable amount (8% to 15%) in all fractions. The content of apo A-II decreased linearly from faster (20%) to slower floating lipoproteins (10%).

Resolution and Identification of HDL Subclasses by Gradient Gel Electrophoresis

Densitometric patterns of Coomassie-stained GGE gels of the ultracentrifugal d < 1.20 g/ml fractions isolated from plasma of six A-I_M^- and ten A-I_M^+ subjects are shown in Figures 3 and 4. The particle size intervals of the major HDL subpopulations encountered in normal plasma are indicated under each of the GGE patterns. These subpopulations have been previously described according to their relationship to the major ultracentrifugal HDL subclasses:[14] three subpopulations occur within the HDL_3 subclass [$(HDL_{3a})_{GGE}$, $(HDL_{3b})_{GGE}$, and $(HDL_{3c})_{GGE}$], and two occur within the HDL_2 subclass [$(HDL_{2b})_{GGE}$, $(HDL_{2a})_{GGE}$]. The peak maxima of the above subpopulations fall within specific particle size intervals that have been used to de-

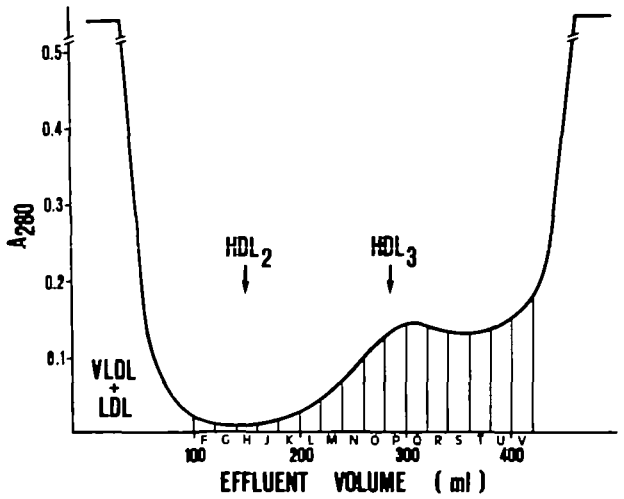


Figure 2. Rate zonal ultracentrifugation (Ti14 rotor) profile of HDL from an A-I_M^+ subject. Twenty-milliliter fractions were collected and analyzed for lipid content and apolipoprotein composition.

Table 3. Apo A-I Isoprotein Distribution in Zonal Ultracentrifugal Fractions Isolated from Three A-I_M^+ Subjects

Fraction	A-I-1	A-I-2	A-I-3
J*	20.1†	43.0	36.8
K	20.2	42.6	37.2
L	23.4	40.2	36.3
M	25.9	37.7	36.5
N	25.5	39.8	34.6
O	27.3	37.5	35.3
P	25.5	37.6	36.9
Q	22.8	42.1	35.2
R	23.4	40.0	36.6
S	23.9	40.4	35.7
T	23.8	41.1	35.3
U	22.1	42.8	35.1

Values are percent of total apo A-I.
*For fraction identification, refer to Figure 2. †The mean of two IEF separations for each subject.

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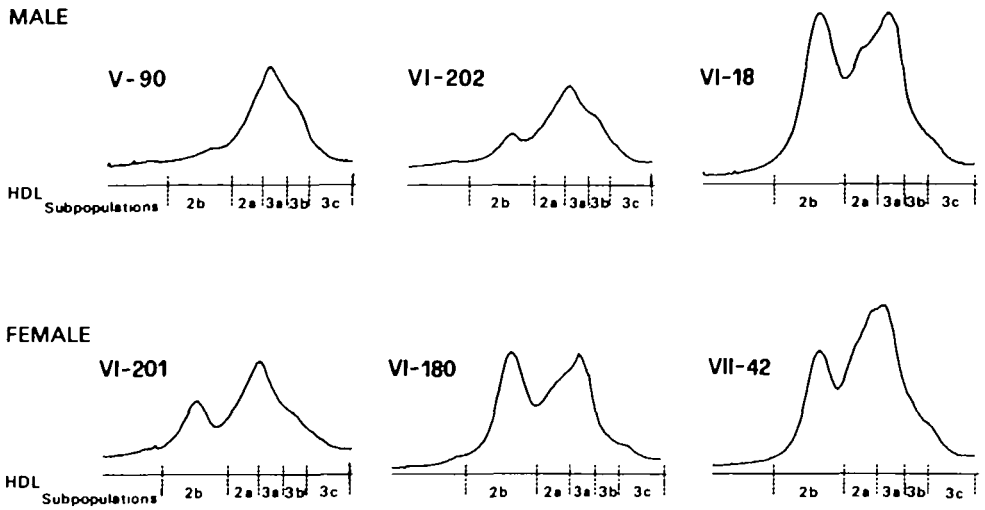


Figure 3. Gradient gel electrophoretic profiles of the ultracentrifugal d < 1.20 g/ml fractions isolated from six A-I_M^- subjects identified by the reported kindred positions.[12] Plasma triglyceride values in the analyzed subjects: V-90 = 89 mg/dl; VI-202 = 72 mg/dl; VI-18 = 119 mg/dl; VI-201 = 196 mg/dl; VI-180 = 58 mg/dl; VII-42 = 87 mg/dl.

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fine the subpopulations: 12.9 to 9.7 nm, $(HDL_{2b})_{gge}$; 9.7 to 8.8 nm, $(HDL_{2a})_{gge}$; 8.8 to 8.2 nm, $(HDL_{3a})_{gge}$; 8.2 to 7.8 nm, $(HDL_{3b})_{gge}$; 7.8 to 7.2 nm, $(HDL_{3c})_{gge}$.

HDL patterns of A-I_M^+ subjects were characterized by major peaks with maxima primarily within the particle size intervals of $(HDL_3)_{gge}$ subpopulations, namely the $(HDL_{3a})_{gge}$ and $(HDL_{3b})_{gge}$. Peaks with maxima in the $(HDL_{2b})_{gge}$ interval, which in normal subjects contains the major HDL_2 subclass, were either minor or nondetectable. Compared to patterns of A-I_M^- HDL, A-I_M^+ HDL were unique in consistently exhibiting a distinct peak within the $(HDL_{3b})_{gge}$ interval.

The mean particle sizes of the major components with peak maxima in the $(HDL_{3a})_{gge}$ and $(HDL_{3b})_{gge}$ intervals in A-I_M^+ (8.6 ± 0.1 and 8.0 ± 0.03 nm, respectively) were almost identical to those in A-I_M^- (8.4 ± 0.07 and 7.9 ± 0.02 nm, respectively). The mean particle size of the major component in the $(HDL_{2b})_{gge}$ interval was, however, significantly smaller (9.9 ± 0.2 vs. 10.3 ± 0.3 nm) in the A-I_M^+ compared with that in the A-I_M^-. Thus, the particle sizes of the major HDL_3 subpopulations in subjects with the apo A-I_M variant were not significantly different from sizes of their counterparts in control subjects.

HDL isolated from the ten A-I_M^+ subjects showed three characteristic patterns (I, II, and III, Figure 4), reflecting the relative contributions of the two major peaks. Pattern I was characterized by a predominant peak in the $(HDL_{3b})_{gge}$ interval, together with a peak of lower amplitude in an interval which includes both $(HDL_{2a})_{gge}$ and $(HDL_{3a})_{gge}$, $[(HDL_{2a+3a})_{gge}]$. In Pattern II, the two peaks were located within approximately the same particle size intervals as in Pattern I but now were of comparable amplitude. Pattern III was identified by a markedly increased amplitude of the peak in the $(HDL_{2a+3a})_{gge}$ interval, together with a distinct peak in the $(HDL_{3b})_{gge}$ interval, as in Patterns I and II.

The group of A-I_M^+ with pattern I exhibited the lowest mean HDL-C value (9.3 ± 1.5 mg/dl) compared with that of A-I_M^+ with Patterns II and III (II: 14.6 ± 4.6 mg/dl; III: 28.0 ± 9.9 mg/dl). Thus, at reduced HDL-C levels, the particle size pattern became relatively enriched in the smaller HDL subpopulations. Progressive, relative enrichment of HDL patterns with species of smaller particle size as a function of decreasing plasma concentration of HDL-C has also been observed in normal subjects from other populations.[22] For normal male adults (age range 35 to 59 years, n = 91), the correlation coefficient relating the percent of the total HDL pattern in the particle size interval of the small subpopulation $(HDL_{3b})_{gge}$ vs. HDL-C was −0.79 ($p < 0.001$).[22] The correlation coefficient between the same parameters for the small sample of ten A-I_M^+ subjects was −0.90 ($p < 0.001$). The mean HDL-C plasma level in the above ten A-I_M^+ subjects was 15.7 ± 8.3 mg/dl, while the mean level in the 91 normal male subjects was 44.0 ± 11.6 mg/dl.

HDL Subpopulations and Plasma Lipids

It is well established that low levels of HDL-C are frequently observed in association with elevated plasma TG levels and this also appeared to be the case for A-I_M^+. Thus, for 29 A-I_M^+ subjects,[12] the correlation coefficient for plasma TG level vs. HDL-C was −0.35 ($p < 0.05$); for the 91 normal males cited above, the correlation coefficient for the same variables was −0.49 ($p < 0.001$). When values of plasma TG and HDL-C were plotted for ten A-I_M^+ and six A-I_M^- subjects whose HDL were analyzed by electrophoresis, the group of A-I_M^+ subjects with Pattern I exhibited the lowest mean HDL-C value and the highest mean TG value, relative to those of A-I_M^+ subjects with Patterns II and III (Figure 5). These data suggest that the Pattern I distribution, in which the relative content of the smaller $(HDL_{3b})_{gge}$ subpopulation was increased, reflected the presence of both elevated plasma TG levels and very low levels of HDL-C. The observation of a relative enrichment of HDL with smaller sized particles in the presence of hy-

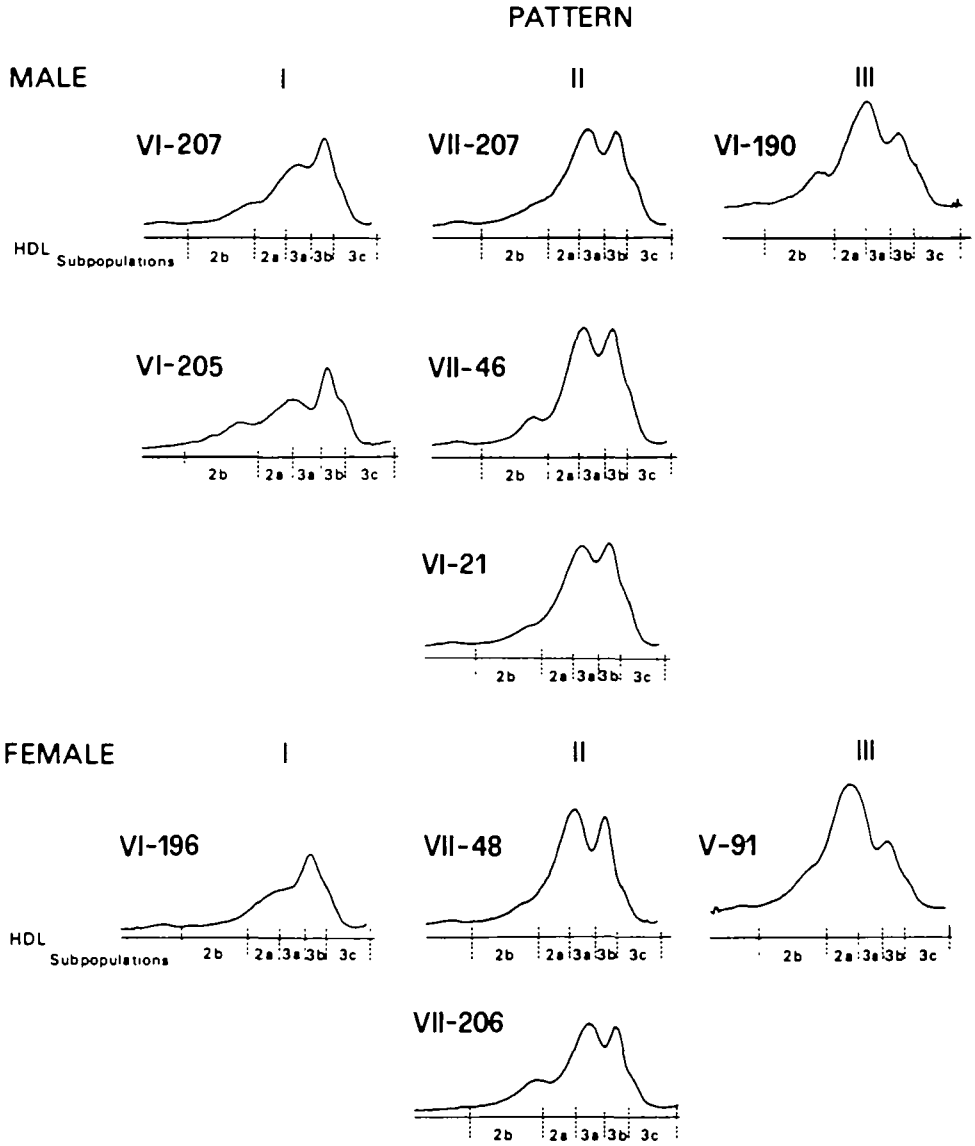


Figure 4. Gradient gel electrophoretic (GGE) profiles of the d < 1.20 g/ml fractions isolated from ten A-I_M^+ subjects identified by the reported kindred position.[11] There are three different patterns (I, II, and III) that reflect the relative contributions of the two major peaks, corresponding to $(HDL_{3a})_{gge}$ and $(HDL_{3b})_{gge}$ subpopulations.

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pertriglyceridemia (222 to 2,500 mg/dl) and reduced HDL-C levels (22 to 30 mg/dl) has been previously reported by Eisenberg et al.[23]

Pattern II, with $(HDL_{3b})_{gge}$ and $(HDL_{2a+3a})_{gge}$ peaks of similar amplitude, was associated with normal triglyceridemia as well as low HDL. Finally, Pattern III, with a markedly increased amplitude of the $(HDL_{2a+3a})_{gge}$ peak, appeared in only two subjects, a male 37 years old actively engaged in marathon running, and a female 66 years old. These two subjects showed normal triglyceridemia and HDL-C levels in the range of 20 to 40 mg/dl.

Analytic Ultracentrifugation

Analytic ultracentrifugal patterns of HDL from six A-I_M^+ subjects, whose plasma levels of HDL-C ranged from 8 to 35 mg/dl, are shown in Figure 6. The total plasma HDL of this group of A-I_M^+ subjects ranged from 19.5 to 261.5 mg/dl and showed a predominance of HDL_3 as gauged by the pattern area within the flotation interval of $F°_{1.20}$ 0 to 3.5. HDL material in the flotation interval of $F°_{1.20}$ 3.5 to 9.0 corresponding to the HDL_2 subclass was observed in A-I_M^+ subjects at the higher plasma HDL-C levels. Plasma concentrations of HDL_2 in these subjects ranged from 0 to 55 mg/dl. At the higher levels of HDL-C, the flotation rate of the major HDL Schlieren peak was also shifted to higher values, consistent with the presence of the additional faster floating HDL_2 species. Analytic ultracentrifugal patterns of HDL (not shown) from the four A-I_M^- subjects with HDL-C and total plasma HDL levels ranging between 42 and 76 mg/dl and 201 and 309 mg/dl, respectively, gener-

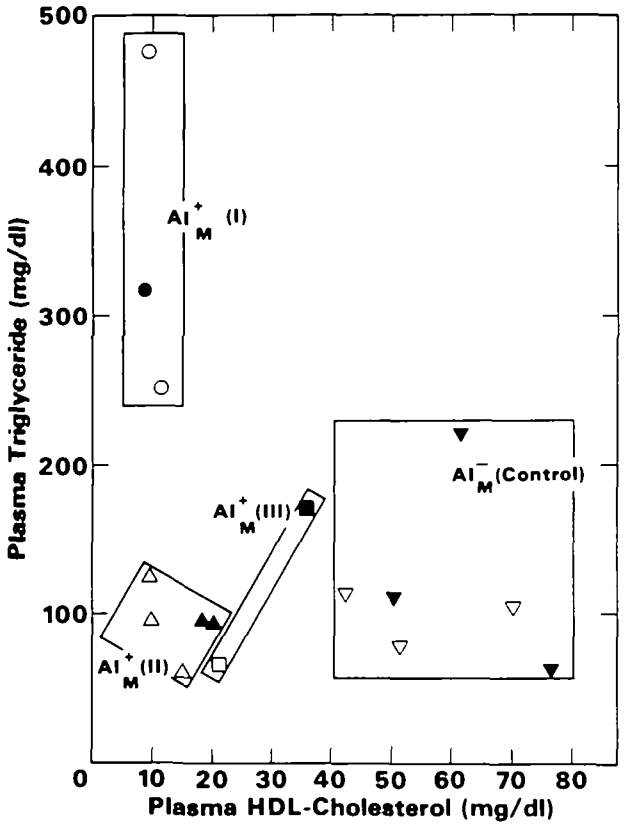


Figure 5. Plot of plasma triglyceride and HDL cholesterol levels in ten A-I_M^+ subjects with different $(HDL)_{GGE}$ patterns (see Figure 4) and in six A-I_M^- subjects (▼▽). Pattern I (●○) is characterized by a predominant peak in the $(HDL_{3b})_{GGE}$ interval; Pattern II (▲△) shows equivalent peaks in the $(HDL_{3b})_{GGE}$ and $(HDL_{2a+3a})_{GGE}$ intervals; Pattern III (■□) is characterized by a relative abundance of particles in the $(HDL_{2a+3a})_{GGE}$ interval. Open symbols = males; closed symbols = females.

ally showed HDL_3 levels higher (164 to 214 mg/dl) than those in A-I_M^+ subjects (25 to 183 mg/dl); the range of HDL_2 in A-I_M^- subjects was 4 to 133 mg/dl. The analytic ultracentrifugal results were consistent with the rate zonal ultracentrifugal data.

Discussion

The isolation and characterization of HDL from a group of carriers of the apo A-I_M variant and from age- and sex-matched controls was achieved by two different techniques. Both rate zonal UCF and GGE confirmed the marked reduction of HDL_2 in the carriers previously detected by cross-linking studies.[10] Peaks with maxima in the HDL_2 region were undetectable in 17 out of the 24 A-I_M^+ subjects analyzed by zonal UCF and in six out of 10 A-I_M^+ subjects tested by GGE. Furthermore, the HDL_2 present in the A-I_M^+ subjects differed from control HDL_2 particles in both composition and size.

Reduced HDL_2 levels have been detected in atherosclerotic patients[24] and in hypertriglyceridemic subjects.[23] The former observation led to the suggestion that HDL_2 might be responsible for the protective effect of HDL against atherosclerosis.[25] The A-I_M^+ subjects, in spite of the markedly reduced levels of HDL, did not suffer from premature cardiovascular disorders.[11] In hypertriglyceridemic subjects, the decreased HDL_2 levels are probably related to the reduced breakdown of TG-rich lipoprotein particles, which normally supply surface material for the direct production of HDL_2 and/or for the conversion of HDL_3 to HDL_2.[26] As a group, the A-I_M^+ subjects showed increased plasma TG levels[11] and generally developed hypertriglyceridemia with aging: the catabolism of TG-rich lipoproteins is, however, apparently normal in these subjects.[6]

Although it cannot be excluded that the apparent absence of HDL_2 in the A-I_M^+ subjects is simply a reflection of the low mass of total HDL,[17] it may be noted that no correlation between HDL_2 and total HDL levels could be detected in the A-I_M^+ subjects; furthermore, in the A-I_M^+ females, who showed higher total HDL concentrations compared to male carriers,[11] the HDL_2/HDL_3 ratio was similar to that recorded in the A-I_M^+ males.

Partial conversion of HDL_3 into HDL_2-like particles occurs in vitro by the uptake of lipids (PL and FC) and apolipoproteins (particularly apo C) released from the surface of lipolysed VLDL.[26] This process, apparently saturable for lipids and apolipoproteins, is not, however, sufficient for the conversion of HDL_3 to HDL_2. Conversion to HDL_2 would probably require cholesterol esterification for further core build-up and some remodeling of lipid and apolipoprotein composition.[27] In the A-I_M^+ subjects, LCAT activity is reduced compared to normal subjects[12] which probably contributes to the increased FC/CE ratio in plasma and particularly in HDL_3 and may hinder conversion to HDL_2. Furthermore, the presence of large apolipoprotein complexes, apo A-I_M–A-II and apo A-I_M–A-I_M, on the surface of HDL may impair remodeling and transfer of apolipoprotein molecules from one particle to another.[28]

Previous studies of hypertriglyceridemic subjects by rate zonal UCF showed that, in the presence of low HDL levels, slower floating HDL_3, also called HDL_{3D}, may be the only HDL population.[13] Similarly, in patients with severe hypertriglyceridemia due to apo C-II deficiency,[29] only very dense, CE-poor and TG-rich HDL_3 can be detected. The studies with A-I_M^+ subjects provided an unusual occasion to examine subjects with generally very low HDL levels but with quite variable triglyceridemia. In HTG A-I_M^+ subjects, HDL_3 of significantly different structure and composition could be detected, compared to NL A-I_M^+. These particles have a lower flotation rate, are richer in TG, and show a higher FC/CE ratio. These data confirm previous observations[13, 23] on the crucial effect of plasma TG levels on the distribution of HDL subclasses in humans.

The GGE studies provided a better differentiation between NL and HTG A-I_M^+ subjects based on the HDL_3 subfraction pattern. The HDL_3 particle distribution in all analyzed A-I_M^+ subjects differed significantly from that of normal subjects, the major modification being the presence of a distinct peak in the $(HDL_{3b})_{GGE}$ interval that was undetectable in controls. The observation of three different patterns, I, II, and III (Figures 4 and 5), indicates that the relative amount of the small $(HDL_{3b})_{GGE}$ particles in the A-I_M^+ subjects is mainly determined by the plasma TG levels, and possibly reflects total HDL in plasma.

Distinct peaks in the small HDL particle size intervals

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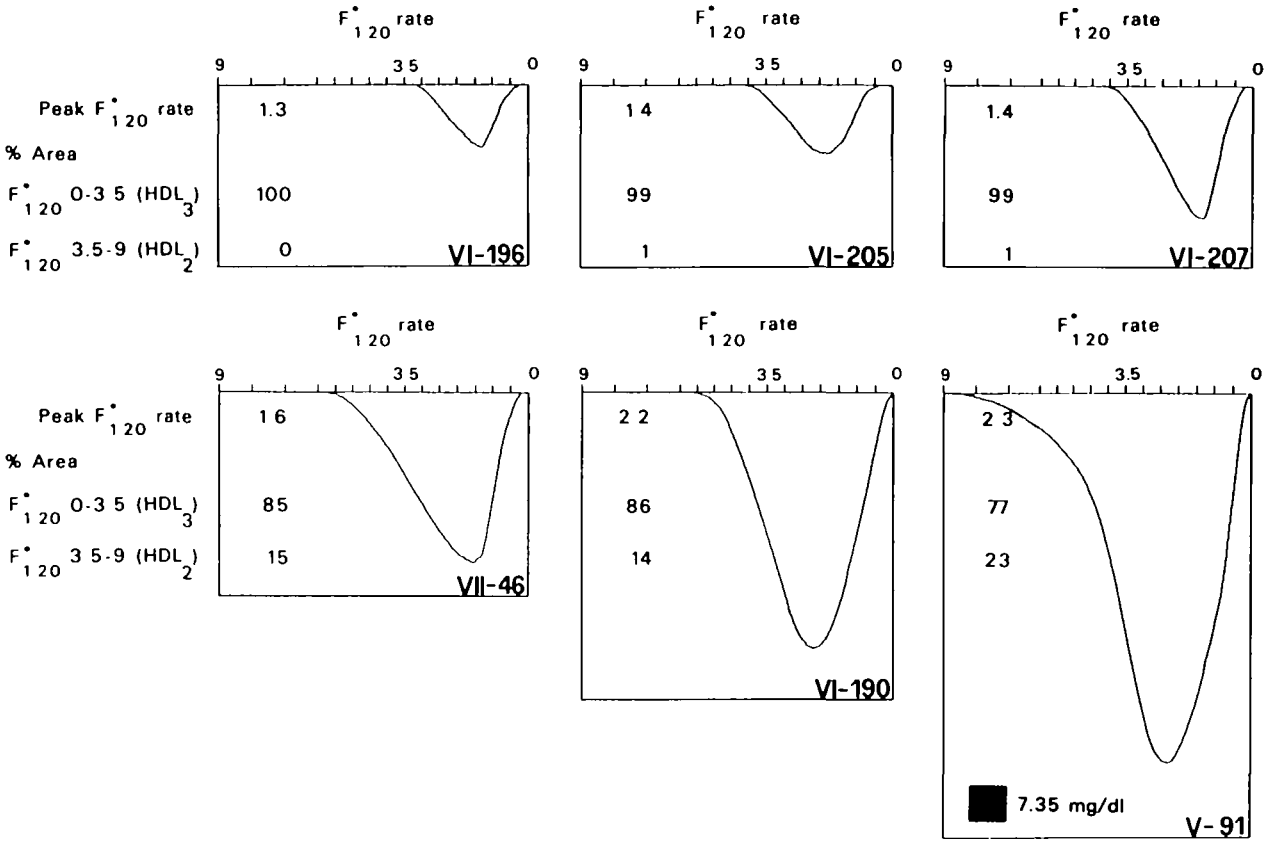


Figure 6. Analytic ultracentrifugal patterns of HDL from six A-I_M^+ subjects. Subjects are identified by the reported kindred position.[11] Total HDL concentration and sex in the examined subjects were: VI-196, F, 19.5 mg/dl; VI-205, M, 21.2 mg/dl; VI-207, M, 51.4 mg/dl; VII-46, M, 96.7 mg/dl; VI-190, M, 151.9 mg/dl; V-91, F, 261.5 mg/dl.

have been detected in several lipoprotein disorders, i.e., hypertriglyceridemia,[14] LCAT deficiency,[30] fish-eye disease,[31] apo A-I–C-III deficiency,[32] and in human cord blood.[33] All these conditions differ from the A-I_M^+ by the presence of larger $(HDL_{2b})_{gge}$ particles and/or even smaller $(HDL_{3c})_{gge}$ particles. The large HDL detected in apo A-I–C-III and LCAT deficiencies, as well as in human cord blood, contain apo E mainly in the form of apo E–A-II complex as a major protein constituent.[30, 32, 33] These particles can accumulate in plasma reflecting a slow catabolism, due to the inability of the apo E–A-II complex to interact with apo B,E receptors.[34] The smaller $(HDL_{3c})_{gge}$, detected in fish-eye disease[31] may represent unreacted precursors of normal HDL.[35] Small, spherical HDL, similar to those secreted by the intestine,[36] have also been detected in subjects with LCAT deficiency.[37] On the other hand, in vitro studies on normal plasma incubated with mercaptoethanol, which stimulates LCAT activity, showed simultaneous conversion of $(HDL_{3a})_{gge}$ and $(HDL_{3b})_{gge}$ to larger subpopulations and to particles in the $(HDL_{3c})_{gge}$ interval.[38] Hepatic lipase is also known to participate in the formation of small HDL_3 particles both in vitro[39] and in vivo[40] through the hydrolysis of the TG core of large, TG-rich HDL_2 particles.

The origin of the $(HDL_{3b})_{gge}$ particles and their relationship to hypertriglyceridemia is less clear. Such particles may be present in HTG subjects with relatively normal HDL levels[14] and with very low HDL levels, as in fish-eye disease,[31] LCAT deficiency,[30] and in the A-I_M^+ subjects. On the other hand, distinct $(HDL_{3b})_{gge}$ peaks have been detected in human cord blood[33] and in apo A-I–C-III deficiency,[32] conditions associated with low plasma TG levels. In the A-I_M^+ subjects, the reduced LCAT activity, which is directly correlated to the total HDL mass,[12] may play a role in the formation of the anomalous HDL particles. The hypertriglyceridemia, which is apparently related to the amount of the mutant apolipoprotein[9] and not inherited as a separate disorder,[11] can also modify the normal conversion of HDL particles in the plasma. In vitro studies on the incubation of A-I_M^+ plasma, both in the presence and absence of active LCAT, are in progress in order to gain more information on the possible origin of the A-I_M^+ HDL particles. It should also be underlined that the need for HDL particles in the A-I_M^+ subjects to accommodate large apolipoprotein complexes on their surface, such as apo A-I_M–A-I_M and A-I_M–A-II, may markedly modify the physical properties of these lipoproteins. In particular, HDL_3 floating in the usual density region, appear to be rich in normal A-I, thus possibly representing normal HDL_3 particles corresponding to the $(HDL_{3a})_{gge}$ subpopulation detected in the plasma of the A-I_M^+ subjects. Slower floating and denser HDL appear to be enriched in the apo A-I_M–A-II complex corresponding to the anomalous lipoproteins migrating in the $(HDL_{3b})_{gge}$ interval.

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The occurrence of hypertriglyceridemia in the A-I_M^+ individuals may be due to the presence of HDL particles which contain only the apo A-I_M–A-II complex, and hence are inactive substrates for the LCAT reaction.[41] The correlation between the relative $(HDL_{3b})_{99e}$ content and triglyceridemia in the A-I_M^+ supports this hypothesis. Reduced levels of acceptors of surface material from VLDL lipolysis may lead to the accumulation of TG-rich lipoproteins in plasma. With reduced CE formation, combined with a marked decrease of the plasma pool of normal HDL, the transfer of CE to VLDL may also be impaired.[42] The exchange of CE and TG will occur predominantly between VLDL and LDL,[43] with consequent formation of TG-rich and CE-poor LDL in plasma.[12] By these mechanisms, A-I_M^+ subjects with HTG may represent the final result of a series of events where a progressive derangement of the complex pathways regulating the interaction between HDL and TG-rich lipoproteins may take place accounting for the appearance of hypertriglyceridemia in the older A-I_M^+ subjects.[11]

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Index Terms: apolipoprotein A-I_{Milano} variant • triglyceridemia • HDL subfractions • rate zonal ultracentrifugation • gradient gel electrophoresis

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